ERD/ERS, MI & ErrP — Training Methodology

Two-model EEG paradigm: 3-class motor imagery (rest / left / right) with the ERD/ERS gate built in, plus error-related potentials

biopipe / Hybrid EEG–EMG BMI • OpenBCI Cyton (8 ch, 250 Hz) + Ultracortex • Montage Fz, FC5, FC6, C3, Cz, C4, P3, P4 (A1 ref, A2 bias) • Revision 2, 2 October 2026 — replaces the 21 September 2026 three-model version (changes listed at the end)

1. Overview

The EEG side trains exactly **two models**:

- **MI model — rest / mi_left / mi_right.** One 3-class classifier. Its *rest* class is the ERD/ERS gate: sensorimotor mu/beta rhythms are strong at rest and desynchronise during imagery, so “rest vs imagery” is the gate and “left vs right” is the command — learned together, with no separate gate model.
- **ErrP model — correct / error.** Detects the brain’s response to a wrong outcome so the controller can veto or undo an action.

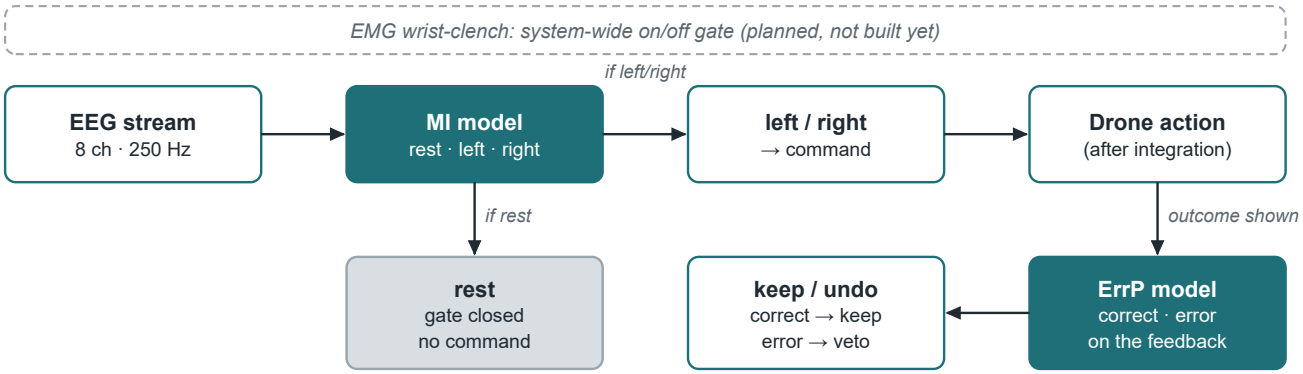


Figure 1 — Control loop. A rest prediction closes the gate; a left/right prediction becomes a command, and the ErrP model checks the brain’s response to its outcome.

2. Montage

Physically available holes only (FCz, FC3/FC4, CP3/CP4 and CPz are unavailable). Load it with **Config → Load MI/ErrP preset**. Standard OpenBCI EEG wiring: electrodes in the bottom-row inputs N1P–N8P, reference ear clip in SRB2, bias ear clip in BIAS.

Table 1 — Electrode map

Electrode	GUI ch	Cyton input	Used for
Fz	0	N1P (ch 1)	ErrP fronto-central (ERN/FRN); blink check
FC5	1	N2P (ch 2)	MI — left premotor/hand area (CSP)
FC6	2	N3P (ch 3)	MI — right premotor/hand area (CSP)
C3	3	N4P (ch 4)	MI — left hand area: ERD during *right*-hand imagery
Cz	4	N5P (ch 5)	MI midline; ErrP central (ERN, Pe)
C4	5	N6P (ch 6)	MI — right hand area: ERD during *left*-hand imagery
P3	6	N7P (ch 7)	MI posterior; ErrP Pe; eyes-closed alpha check
P4	7	N8P (ch 8)	MI posterior; ErrP Pe; eyes-closed alpha check
A1 (left ear)	ref	SRB2	Reference — required
A2 (right ear)	bias	BIAS	Bias (driven ground) — required

3. Shared EEG settings (Config and Monitor tabs)

These apply to both models. Notch, band limits and common-average reference change only what the Monitor shows: recordings are saved **raw**, and each model applies its own band-pass when it trains and runs (Appendix A).

Table 2 — Config and Monitor settings

Tab	Setting	Recommended	Options / range	Affects
Config	Profile	One per setup, e.g. *MI-8ch*	Load · Delete · Save as	Restores every Config setting below in one click.
	Source	cyton (synthetic for a dry run)	cyton · lsl · synthetic	Where samples come from.
	Cyton port	The dongle's port	e.g. COM3	Device Manager → Ports. Close the OpenBCI GUI first.
	Sample rate	250	Hz	Must match the board (Cyton: 250 Hz).
	Montage (10-20 head)	Load MI/ErrP preset	Click sets Data · Ref (SRB) · Bias · Clear; channel 0–7 per Data node	Electrode→channel map saved with every recording. Live and pooling need the same map.
	Actions	mi_left, mi_right	Add · Remove	The two MI classes (rest is added automatically). Names containing "left"/"right" set the cue arrows. Keep exactly these two.
	Notch Hz	50	Hz	Monitor only (India mains).
	Band low / high	1 / 40 (high 45–50 to see γ)	Hz	Monitor only.
	Common-avg ref	On	On · off	Monitor only.
	Window ms	1000–2000	ms	Monitor band-power window.
	Step ms	250	ms (Live minimum 150)	Monitor refresh and Live update rate.
	Alpha thresh	≈ 0.2, then calibrate	0–1 (alpha share)	Alpha light; electrodes set to the most posterior pair (P3/P4) on Apply.
	Apply & Save	After any change	—	Writes config/eeg.yaml and stops the stream — press Start again.
Monitor	alpha test	On	Checkbox	Lights when P3/P4 alpha share exceeds Alpha thresh.
	waterfall	On during checks	Checkbox	Spectrogram: frequency across (0 → Band high), time downward.
	simulate eyes-closed	—	Synthetic source only	Injects alpha to test the pipeline without hardware.

4. Headset setup and signal quality (every session)

- **Connect reference and bias.** A1 → SRB2, A2 → BIAS (never AVDD/AVSS/AGND). Without both you record mains hum and drift, not EEG — the giveaway is a band composition swamped by δ.
- **Contact.** Part the hair and twist each comb down to the scalp. Aim for impedances in the low tens of kΩ and similar across channels; mismatched or very high contacts let mains leak in, and common-average reference spreads one bad channel into all of them.
- **Monitor checks** (EEG → Monitor → Start): no channel flagged **RAILED**, **FLAT** or *near rail* — judge by the flags, not the traces: with common-average reference on, a dead channel still draws a convincing trace. A blink gives a slow deflection at Fz; a jaw clench gives a broadband burst on all channels that settles back; the band composition shows a spread of θ/α/β rather than a wall of δ.
- **Eyes-closed test.** 15 s eyes open, 15 s closed: alpha should rise at P3/P4 (alpha light, or a bright ~10 Hz column on the waterfall). Parietal alpha is weaker than occipital, so set Alpha thresh between your eyes-open and eyes-closed values.
- **Environment.** Upright and relaxed, screen at eye level ~60 cm; unplug the laptop charger if possible; away from mains cables and motors; phone away; no gum.
- **Dongle latency (Windows).** FTDI latency timer → 1 ms (Device Manager → COM port → Port Settings → Advanced).

5. Recording rules (both models)

- Stay still and silent during recorded periods — no talking, head movement or shifting.
- Blink only during the yellow *get ready* cue (MI) or the fixation cross (ErrP) — never during REST, imagery or an outcome.
- No jaw clenching or facial/neck tension: muscle activity swamps EEG.
- Keep a soft, steady gaze on the cue. Ignore the small countdown in the corner — it is deliberately dim and off-centre.
- Runs of about 3–5 minutes with breaks; fatigue degrades both signals.

6. Model 1 — Motor imagery (rest / left / right)

6.1 Why rest is a class

Imagining a hand movement desynchronises the mu (8–12 Hz) and beta (13–30 Hz) rhythms over the opposite hemisphere's hand area — C3 for the right hand, C4 for the left — and they rebound (ERS) afterwards [1, 2]. During rest they stay high. Training *rest* as its own class therefore teaches the ERD/ERS gate and the left/right decision in one model; P(rest) acts as a soft gate.

6.2 Subject technique

- **Imagine, don't move.** Imagine the *feeling* of slowly squeezing the cued hand, as if your own muscles did it — not a video of yourself. Hands stay relaxed and still. A real squeeze is motor execution (another signal plus muscle artefact), so a model trained on it fails on pure imagery.
- **Warm-up (not recorded):** 3–5 slow real squeezes per hand, then reproduce that feeling without moving.
- Use the **same strategy every trial** — consistency beats intensity.
- **Yellow** = prepare, don't start. **Green** = imagine for the whole block; stop when the cue changes. **Blue REST** = relax, fixate and keep still — rest is recorded data too.

6.3 Recording protocol

Each rep records REST, LEFT and RIGHT once each **in random order**, each after an unrecorded 2 s *get ready* cue; counts stay equal and no cue repeats back-to-back. Each recorded block is one **trial**; its first 0.4 s is dropped as cue-onset transient. A rep takes 18 s at 4 s per class, so 10 reps is a 3-minute run.

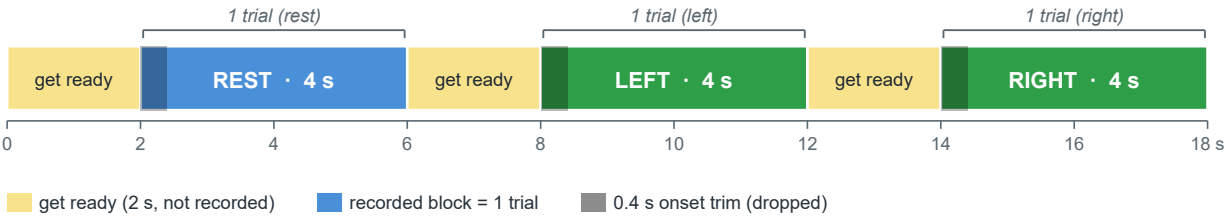


Figure 2 — One example MI rep at the default 4 s per class (colours as on the Cue display). The order inside each rep is randomised.

How much: aim for at least 25–30 trials per class — for example 3 runs of 10 reps. Cross-validation uses up to 5 folds; with fewer than 5 trials in a class it uses fewer folds, and with fewer than 2 it cannot validate at all.

6.4 Features and model

Raw window → band-pass 8–30 Hz → Common Spatial Patterns (CSP) → log-variance → standardise → classifier. CSP learns spatial filters that maximise the variance difference between classes [3]. With three classes it runs one-vs-rest — a CSP per class against the other two, 4 filters each — giving 12 features [4].

Table 3 — Classifier choices (Train → Model)

Classifier	Type	When to use
lda	Linear discriminant (default)	Start here: robust with little data, fast; the standard MI choice [5].
svm	Linear support-vector machine	Often matches LDA; may edge ahead with more trials.
logreg	Logistic regression (linear)	Similar boundary to LDA; well-behaved probabilities.
rf	Random forest, 200 trees	Non-linear; needs more data to beat the linear models.
tree	Single decision tree	Interpretable but tends to overfit CSP features — use for comparison only.

What to expect: chance is 33%. With dry electrodes an early target is roughly 50–70% balanced accuracy, with rest vs imagery usually separating more easily than left vs right. A sizeable minority of users need several sessions before their imagery patterns separate [6] — treat these figures as indicative, not a guarantee.

6.5 Settings

Table 4 — MI settings in the GUI

Tab	Setting	Recommended	Options / range	Notes
Record	Subject	Your ID, e.g. *amogh*	Free text	Names the folder eeg_<subject>_<date-time>.
	Paradigm	MI (left/right)	MI · ErrP	REST, LEFT and RIGHT blocks (3-class).
	Seconds/class	4	2–30 s	Length of each recorded block; must exceed the Train window + 0.4 s.
	Reps	10 per run	1–15	Each class once per rep, random order.
	Description	Run notes	Optional	Saved in the recording's JSON.
Train	Modality · Paradigm	EEG · MI (rest/L/R)	—	Lists only MI recordings.
	Model	lda	lda · svm · logreg · rf · tree	See Table 3.
	Win / Step ms	1000 / 250	ms	Training window and hop; also the model's Live window. Longer is steadier but slower to respond.
	Recordings	2–3 runs, pooled	Ctrl/Shift-click; Refresh	Same montage, rate and labels required.
	Description	Model notes	Optional	Saved in the model's JSON.
Live	Models	The saved MI model	Multi-select; refresh; Start	Window from the model; update rate from Config Step.

6.6 In the app

1. **Config** → Load MI/ErrP preset → Source *cyton* + COM port → Apply & Save.
2. **Monitor** → run the checks in section 4.
3. **Record** → Subject → Paradigm *MI (left/right)* → Seconds/class 4, Reps 10 → Description → Record session, and follow the cue. Repeat for 3 runs with breaks.
4. **Train** → EEG → *MI (rest/L/R)* → Model *lda* → Win/Step 1000/250 → Ctrl-click the runs → Train → read the report → Save model.
5. Optionally retrain the same runs with svm, logreg, rf or tree and keep the model with the best macro-F1.

7. Model 2 — Error-related potentials (correct / error)

7.1 Subject technique

- **Care about the outcome.** Press ← or → toward the blue box, watch where the cursor goes, and let the natural "that's wrong" reaction happen.
- One finger only; never move, sigh or clench on an error — a movement time-locked to errors is the fastest way to ruin the data, because the model learns the artefact.
- Don't try to predict outcomes; react honestly. Blink during the fixation cross.

7.2 Recording protocol

Cursor task (recommended): fixation, then a target box on the left or right; you press ←/→ toward it and the cursor moves — the opposite way at the set error rate. Your own command going wrong evokes an *interaction* ErrP, much stronger than watching [11]; on our data only this task was decodable. A key away from the target, or none within Max wait, leaves the trial unlabelled and it is repeated. **Arrow task** (observation): a target arrow for 0.9 s, then the outcome arrow. In both, right and wrong outcomes look identical except for the direction, the outcome is time-stamped as *correct* or *error*, and each event yields one epoch — one trial for cross-validation.

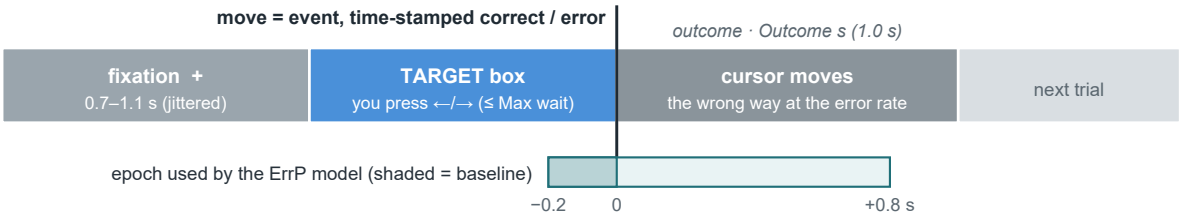


Figure 3 — One cursor-task trial and the epoch extracted around the move.

How much: about 3 s per trial, so 100 trials take roughly 5 minutes. Aim for at least 200 trials (about 60 errors at 0.30); record runs of one task only and train on them together.

7.3 Features and model

Band-pass 1–10 Hz keeps the slow error components — the error-related negativity at fronto-central sites, then the error positivity (Pe) [7, 8]; each −0.2 to +0.8 s epoch is baseline-corrected on −0.2 to 0 s. On this headset ~95% of the 1–10 Hz power is common to all channels (the ear reference's own activity) and buries the error response, so the model filters spatially first: xDAWN [12] learns two filters per class that maximise the evoked response; each epoch is stacked under the two filtered class averages, its shrunk covariance is mapped to the Riemannian tangent space [13], and a balanced logistic regression gives P(error) — the BCI Challenge @ NER 2015 winning approach.

What to expect: on the 28 Oct cursor sessions (180 trials) 0.78 balanced accuracy (label-shuffle p = 0.01) and 0.73–0.81 on a held-out session, peaking at Cz; the arrow task stays at chance, and the old time-bin + LDA model scored 0.55 on the cursor data.

7.4 Settings

Table 5 — ErrP settings in the GUI

Tab	Setting	Recommended	Options / range	Notes
Record	Subject	Your ID	Free text	As for MI.
	Paradigm	ErrP (cursor task)	MI · cursor · arrow	Cursor = press ←/→ (interaction); arrow = watch.
	Trials · Max wait · Outcome s	100 · 3 · 1.0	20–400 · 1–10 · 0.8–5	About 3 s per trial; Max wait is the cursor task only.
	Error rate	0.30	0.10–0.50 (step 0.05)	Share of wrong outcomes; rarer errors evoke a stronger response, but fewer epochs.
	Description	Run notes	Optional	Saved in the recording's JSON.
Train	Modality · Paradigm	EEG · ErrP	—	Lists only ErrP recordings.
	Model · Win/Step	— (hidden)	Fixed	xDAWN + Riemann + LR on −0.2 to +0.8 s epochs.
	Recordings	2+ runs of one task	Ctrl/Shift-click; Refresh	Same pooling rule as MI.
ErrP Live	Test · Trials · Error rate · Threshold	Cursor · 40 · 0.30 · 0.50	Cursor / Arrow / Facts; threshold 0.05–0.95	Each outcome scored; threshold re-scores all. Max wait / Outcome s as in Record.

7.5 In the app

1. Config preset and Monitor checks as for MI.
2. **Record** → Subject → Paradigm *ErrP (cursor task)* → Trials 100, Error rate 0.30 → Record session, and press ←/→ toward the blue box. Repeat.
3. **Train** → EEG → *ErrP* → Ctrl-click the cursor runs → Train → Save model.

8. Evaluation: how to read the report

Training never splits windows at random. Neighbouring windows from the same block are nearly identical, so a random split leaks test data into training and inflates the score [10]. Instead:

- **Grouping:** every MI block and every ErrP event is one trial, and all of a trial's windows or epochs stay in the same fold.
- **Folds:** StratifiedGroupKFold with k = min(5, fewest trials in any class); classes are balanced across folds.

- **Report:** out-of-fold predictions are pooled into the per-class report shown in the Train tab's right-hand box; the saved model is then refit on all the data.

Table 6 — Metrics in the report

Metric	What it means	What to look for
precision	Correct share of a class's predictions.	Low for left/right → rest misread as a command.
recall	Share of a class's windows caught.	MI rest: gate reliability. ErrP error: errors caught.
F1	Harmonic mean of precision and recall.	Compare classifiers by macro-F1 (mean over classes).
support	Windows (MI) or epochs (ErrP) per class.	~11 per 4 s block; if uneven, record more of that class.
balanced accuracy ± std	Mean per-class recall; mean ± std over folds.	Chance 33% (MI) / 50% (ErrP); large std → record more.

Within-session cross-validation is still optimistic. The honest test is a new session — re-seat the headset on another day and run the saved model in Live.

9. Live use

- EEG → Live → refresh → select the MI model → Start. Bars show P(rest), P(left) and P(right), with the top class highlighted and repeated on the DECISION line.
- *Rest* = gate closed, no command. Live currently reports the top class at every update; a confidence threshold and smoothing for drone commands will be tuned during integration.
- The current Config montage must contain the model's electrodes (the preset guarantees this). The window comes from the model and the update rate from Config Step (minimum 150 ms).
- A **data delay** readout (Monitor, Live, ErrP Live) shows how stale the newest sample is and warns when data arrives in bursts; training and live use the same causal filters, so live results match training exactly.
- ErrP models are event-locked, so they are tested on the **ErrP Live** tab instead: it runs the cursor or arrow task, classifies each outcome ~0.8 s after it and scores it against what was shown (accuracy, errors caught, false alarms, confusion matrix).
- ErrP Live's **True / false facts** test shows easy statements (100 true, 100 false) for *Fact time* (default 15 s, 2–60 s); every 0.1 s epoch is classified while one is shown and its *Score* (peak or mean P(error)) is thresholded. Untick *Show results* to hide the verdicts from the subject.

10. Session cadence and pooling

- Short runs with breaks; two sessions on different days beat one long session.
- Retrain each session: EEG shifts between sessions and headset placements.
- Pool runs only when montage, sample rate and labels match — the Train tab blocks mismatches. Trial IDs stay unique across pooled runs.
- Record MI and ErrP as separate sessions.

11. Pre-flight checklist

- ☐ A1 → SRB2 and A2 → BIAS clipped on and wired.
- ☐ Combs on the scalp; contacts low and even.
- ☐ Config: MI/ErrP preset, Source cyton + COM port, Apply & Save.
- ☐ Monitor: no RAILED/FLAT flags, blink visible at Fz, no wall of δ, alpha rises at P3/P4 with eyes closed.
- ☐ Charger unplugged or away from mains; phone away; seated upright.
- ☐ Record: Subject entered, paradigm chosen, description added.
- ☐ MI: imagine — don't squeeze; same strategy; still during REST; blink only on yellow.
- ☐ ErrP: care about the outcome; don't move on errors.

Changes since revision 1

- ERD/ERS is no longer a separate model — it is the MI model's *rest* class.
- MI is 3-class with a selectable classifier (lda, svm, logreg, rf, tree).
- MI cues are block-randomised: equal rest/left/right counts, never the same cue twice in a row.
- *Rest* blocks are recorded and trained, so they must be as clean as imagery blocks (no blinking or fidgeting).
- Evaluation is trial-grouped, class-stratified k-fold cross-validation with per-class precision, recall, F1 and support.
- Recordings carry a *trial* column; the subject is entered in the Record tab.
- New ErrP Live tab: test a trained ErrP model on live EEG with per-trial scoring.
- Causal filtering in training and live: live results match training exactly, and the ErrP verdict arrives ~0.8 s after the outcome.
- Outcomes are mapped to the EEG through a stream clock, immune to USB buffering bursts.
- Reference (SRB) and bias must be physically connected; the alpha check runs on P3/P4.
- Rev. 3: ErrP uses the cursor (interaction) task; outcomes no longer differ in colour or text.
- Rev. 3: the ErrP model is xDAWN + Riemannian tangent space + logistic regression.

Appendix A — Settings fixed in code (not in the GUI)

Table 7 — Fixed processing and timing

Item	Value	Where
MI band-pass	8–30 Hz, causal (same filter live and in training)	eeg_training.MI_BAND
CSP	4 filters per CSP; one-vs-rest for 3 classes (12 features)	eeg_models.MIModel
Onset trim	0.4 s at the start of each recorded MI block	transient_ms in config/eeg.yaml (no GUI field)
MI get-ready cue	2 s, not recorded	Record tab (MI)
MI cue order	Block-randomised: each class once per rep; no immediate repeats	eeg_tab.mi_cue_order
ErrP band-pass	1–10 Hz, causal (same filter live and in training)	eeg_training.ERRP_BAND
ErrP epoch / baseline	−0.2 to +0.8 s / −0.2 to 0 s	ERRP_TMIN, ERRP_TMAX
ErrP features	xDAWN 2 + 2 filters; 12 × 12 covariance → tangent space (78)	eeg_models.ErrPModel
ErrP classifier	Logistic regression, balanced classes	eeg_models.ErrPModel
ErrP task timing	Fixation 0.7–1.1 s · target until the key (arrow: 0.9 s) · outcome 1.0 s	Record tab (ErrP), ErrP Live
Cross-validation	StratifiedGroupKFold, k = min(5, fewest trials in any class); final model refit on all data	eeg_training._grouped_cv
Live MI	Continuous causal 8–30 Hz filter → newest window → model; update every ≥ 150 ms	eeg_tab.EEGEngine
Event timing	Outcome painted, then time-stamped and mapped to its sample via the stream clock	eeg_tab.EEGEngine.index_at
ErrP Live verdict	As soon as the epoch ends (~0.8 s after the outcome)	eeg_tab.EEGEngine

References

[1] Pfurtscheller G, Lopes da Silva FH. Event-related EEG/MEG synchronization and desynchronization: basic principles. *Clin Neurophysiol*. 1999;110(11):1842–1857.

[2] Pfurtscheller G, Neuper C. Motor imagery and direct brain–computer communication. *Proc IEEE*. 2001;89(7):1123–1134.

[3] Ramoser H, Müller-Gerking J, Pfurtscheller G. Optimal spatial filtering of single trial EEG during imagined hand movement. *IEEE Trans Rehabil Eng*. 2000;8(4):441–446.

[4] Dornhege G, Blankertz B, Curio G, Müller K-R. Boosting bit rates in noninvasive EEG single-trial classifications by feature combination and multiclass paradigms. *IEEE Trans Biomed Eng*. 2004;51(6):993–1002.

[5] Lotte F, Bougrain L, Cichocki A, et al. A review of classification algorithms for EEG-based brain–computer interfaces: a 10 year update. *J Neural Eng*. 2018;15(3):031005.

[6] Vidaurre C, Blankertz B. Towards a cure for BCI illiteracy. *Brain Topogr*. 2010;23(2):194–198.

[7] Chavarriaga R, Millán JdR. Learning from EEG error-related potentials in noninvasive brain–computer interfaces. *IEEE Trans Neural Syst Rehabil Eng*. 2010;18(4):381–388.

[8] Chavarriaga R, Sobolewski A, Millán JdR. Errare machinale est: the use of error-related potentials in brain–machine interfaces. *Front Neurosci*. 2014;8:208.

[9] Blankertz B, Lemm S, Treder M, Haufe S, Müller K-R. Single-trial analysis and classification of ERP components — a tutorial. *NeuroImage*. 2011;56(2):814–825.

[10] Varoquaux G, Raamana PR, Engemann DA, et al. Assessing and tuning brain decoders: cross-validation, caveats, and guidelines. *NeuroImage*. 2017;145:166–179.

[11] Ferrez PW, Millán JdR. Error-related EEG potentials generated during simulated brain–computer interaction. *IEEE Trans Biomed Eng*. 2008;55(3):923–929.

[12] Rivet B, Souloumiac A, Attina V, Gibert G. xDAWN algorithm to enhance evoked potentials: application to brain–computer interface. *IEEE Trans Biomed Eng*. 2009;56(8):2035–2043.

[13] Congedo M, Barachant A, Bhatia R. Riemannian geometry for EEG-based brain–computer interfaces; a primer and a review. *Brain–Computer Interfaces*. 2017;4(3):155–174.